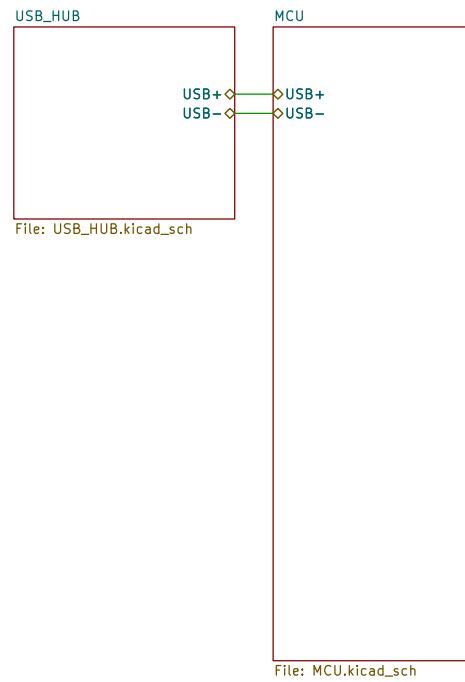
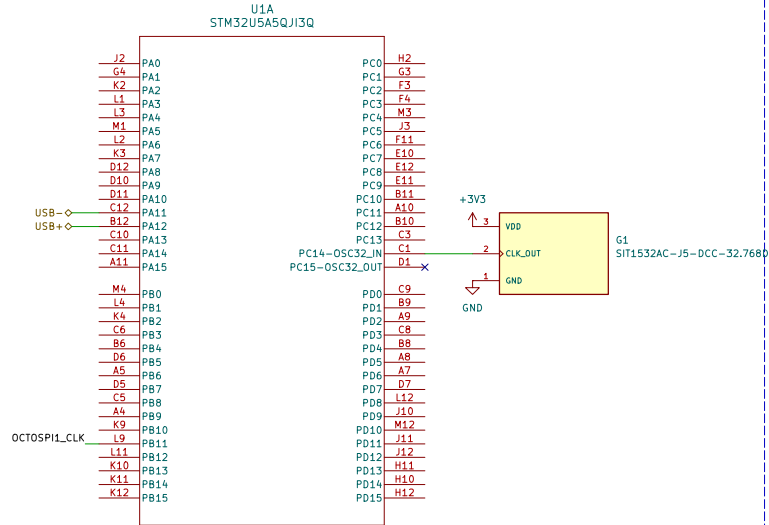
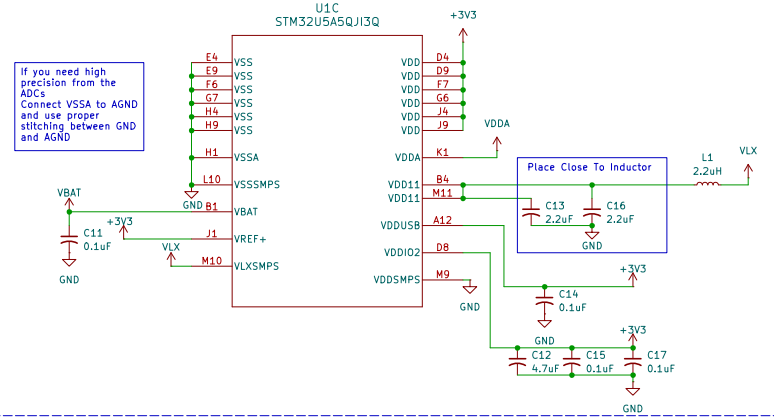


GPIO

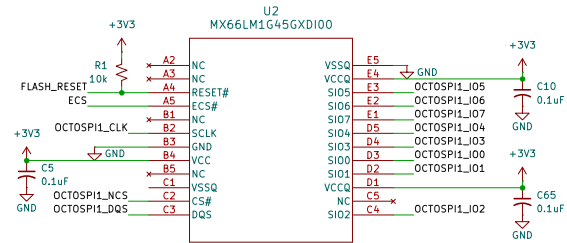


Power

VBAT is currently left
unconfigured, please fix when
power supply is done



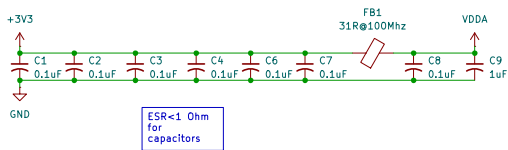
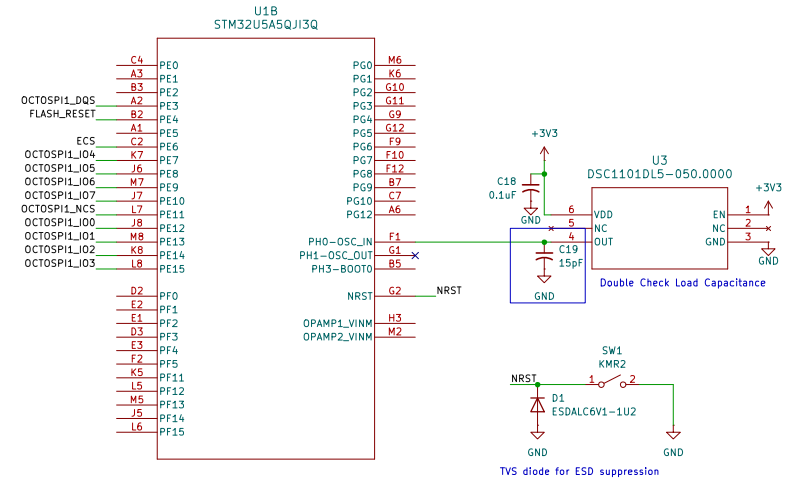
OCTOSPI Flash



Keeping the data lines (I00-I07) on the same physical port (Port E) ensures the internal silicon routing and GPIO toggle timings are as closely matched as possible. Route DQS exactly like a data line. Match the lengths of I0[0:7], CLK, and DQS to within ±2mm of each other to prevent timing skew during high-frequency operations. Ensure all 11 signals are routed over a continuous, unbroken ground plane on the adjacent layer to provide a clean return path and maintain the target single-ended impedance (typically 50Ω).

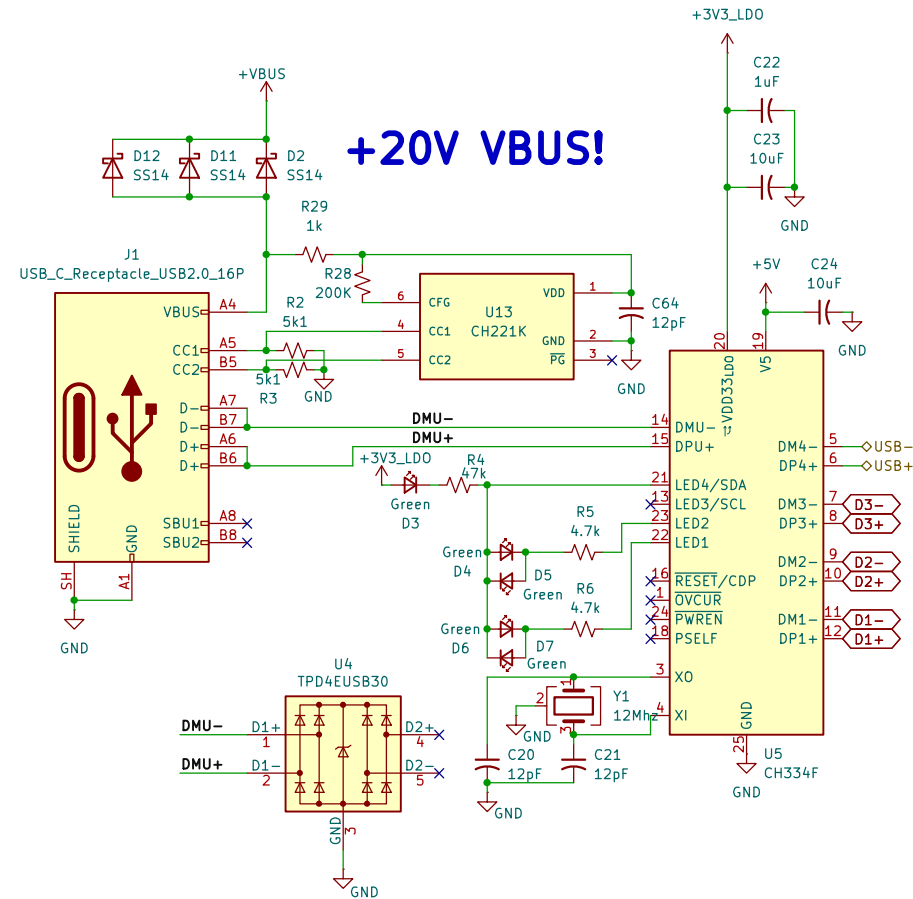
ECS pin will report an error corrected signal, used to log flash degradation
Reset should generally not be externally toggled by the MCU but in the event of a hard reset or crash it can prove useful

GPIO + CLK + RESET



Power Supply Decoupling

Under No circumstances should the LEDs or Load Capacitors be made DNP,
Doing so breaks the USB enumeration process on Windows 11 and stops the hub working entirely specifically on dell laptops,
Donot believe the datasheet
This setup has been validated and works.



May not work with some
USB-PD sources that
use the D- and D+ pins

Standard USB 2.0 Spec routing

Charger IC

Ideal Diodes for reverse polarity protection and power path selection

